

SAFETY DATA SHEET

EMERGENCY CALL: +1 703-741-5970



1. IDENTIFICATION

PRODUCT NAME: Marshal Imazapic

PRODUCT CODES: SKU:D00001756/ D00001757 UVP:DU00000147

Specification: 102D00000403

EPA Reg. No.: 66222-141-101563

COMPANY IDENTIFICATION:

Environmental Science U.S. LLC.

5000 Centregreen Way, Suite 400

Cary NC 27513

2. HAZARD IDENTIFICATION

WARNING

Causes eye irritation (H320)

Harmful if inhaled (H332)

Toxic to aquatic life with long lasting effects (H411)



HAZARD CLASSIFICATION

Health Hazard

Eye Damage / Irritation

Acute Toxicity, Inhalation

Category

2B

4

Physical Hazards

None

Category

-

Environmental Hazards

Hazardous to aquatic environment, long-term

Category

2

HAZARDS NOT REQUIRING CLASSIFICATION

None.

PRECAUTIONARY STATEMENTS

Wash hands thoroughly after handling. Avoid breathing mist/vapors/spray. Use only outdoors or in a well-ventilated area. (P264+P261+P271)

Avoid release to the environment not in accordance with the product label. (H273)

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/attention. (P305+ P351+P338+P337+P313)

IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a Poison Control Center or doctor if you feel unwell. (P304+P340+P312)

Collect spillage. (H391)

Dispose of contents / container in accordance with local regulations. Refer to the product label for specific disposal instructions. (P501)

3. COMPOSITION / INFORMATION ON INGREDIENTS

Common Name

Ammonium Salt of Imazapic

Chemical Name

(±)-2-[4,5-dihydro-4-methyl-4-(1-methylethyl)-5-oxo-1H-imidazol-2-yl]-5-methyl-3-pyridinecarboxylic acid

CAS #

104098-49-9

Composition

23.3%

4. FIRST AID MEASURES

Have the product container or label with you when calling a poison control center or doctor or going for treatment. You may also contact CHEMTREC at 1-800-424-9300 for emergency medical treatment information.

IF IN EYES: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after the first 5 minutes, then continue rinsing eye. If eye irritation persists: Get medical advice/attention.

IF INHALED: Move person to fresh air and keep comfortable for breathing. If person is not breathing, call 911 or an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a doctor or poison control center if you feel unwell.

5. FIREFIGHTING MEASURES

Flash point (TCC): > 100°C (> 212°F)

Fire and Explosion Hazards: Thermal decomposition may release irritating gases and fumes (see Section 10).

Extinguishing Medium: Foam, dry chemical, CO₂ or water fog.

Fire Fighting Equipment: Firefighters should be equipped with self-contained positive pressure breathing apparatus and full bunker gear.

Fire Fighting Instructions: Evacuate area and fight fire upwind from a safe distance to avoid possible hazardous fumes and decomposition products. Dike runoff and do not allow runoff to enter sewers, storm drains or waterways. Foam and dry chemical extinguishing systems are preferred to prevent environmental damage from excessive water runoff.

NFPA Ratings: Health – 2 / Flammability – 1 / Reactivity - 0

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions: Isolate area and keep unnecessary and unprotected personnel from entering. Wear suitable personal protective clothing and equipment as described in Section 8 of this document. Extinguish sources of ignition nearby and downwind and ensure adequate ventilation.

Environmental Precautions: Do not discharge into soil / subsoil or into drains / surface water / groundwater.

Large Spills: Dike spillage and recover and retain as much free liquid as possible for reuse. Pick up remainder with suitable absorbent material. Place into suitable containers for reuse or disposal in a licensed facility. After removal, thoroughly clean contaminated area with water. Collect wash water for approved disposal.

7. HANDLING AND STORAGE

Handling: Use only outdoors or in a well-ventilated area and avoid breathing spray mist. Avoid contact with skin, eyes, or clothing. Wash thoroughly with soap and water after handling. Wash hands before eating, drinking, chewing gum, using tobacco, or using the toilet. Wear appropriate personal protective clothing and equipment (see Section 8 below). Remove clothing immediately if pesticide gets inside. Then wash thoroughly and put on clean clothing. As soon as possible, wash thoroughly and change into clean clothing.

Storage: Keep out of reach of children and animals. KEEP FROM FREEZING (do not store below 20°F). Protect from temperatures above 104°F. Protect containers from physical damage.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Engineering Controls: Facilities storing or utilizing this material should be equipped with an eyewash station and a safety shower.

Pesticide Applicators and Workers: Refer to the product label attached to the product.

Engineering Controls: Workplace should be equipped with a shower and eye-wash station.

Respiratory Protection: If ventilation is inadequate, wear a NIOS-certified (or equivalent) TC23C chemical / mechanical type filter system to remove a combination of particles, gas and vapors.

Personal Protective Equipment (PPE):

Applicators and other handlers must wear: Long-sleeved shirt and long pants, Chemical-resistant gloves made of any waterproof material, Shoes plus socks

Follow manufacturer's instructions for cleaning/maintaining PPE. If no such instructions for washables, use detergent and hot water. Keep and wash PPE separately from other laundry.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance:	yellow-green liquid	pH:	6.0-7.0
Odor:	faint	Kinematic viscosity:	not available
Melting/freezing point:	not available	Solubility:	soluble
Boiling point/Boiling range:	not available	Partition coefficient:	not available
Flammability:	not available	Vapor pressure:	not available
Flammability limits (upper/lower):	not available	Density:	1.101 g/mL (9.19 lbs/gal)
Flash point:	not available	Relative vapor density:	not available
Auto-ignition temperature:	not available	Particle characteristics:	(solids only)
Decomposition temperature:	not available		

10. STABILITY AND REACTIVITY

CONDITIONS TO AVOID: All sources of open flame, heat and spark. Extreme temperatures.

CHEMICAL STABILITY: Stable under all normal use and storage conditions. May thermally decompose.

HAZARDOUS DECOMPOSITION PRODUCTS: Thermal decomposition may generate oxides of carbon and nitrogen.

INCOMPATIBILITY WITH OTHER MATERIALS: Oxidizing agents.

POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION

ORAL TOXICITY (rat LD₅₀): > 5,000 mg/kg

DERMAL TOXICITY (rabbit LD₅₀): > 5,000 mg/kg

INHALATION TOXICITY (rat LC₅₀): > 2.38 mg/L

EYE IRRITATION: Moderately irritating

SKIN IRRITATION: Not irritating

SKIN SENSITIZATION: Not a sensitizer

CARCINOGENICITY:

EPA: Not listed.

ACGIH: Not Listed

NTP: Not Listed

IARC: Not Listed

OSHA: Not Listed

MUTAGENIC TOXICITY: Little evidence of mutagenic effects during *in vivo* and *in vitro* assays.

REPRODUCTIVE TOXICITY: No evidence in animal studies.

12. ECOLOGICAL INFORMATION

Do not apply directly to water, to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when disposing of equipment washwaters. Do not contaminate water used for irrigation or domestic purposes.

This chemical demonstrates the properties and characteristics associated with chemicals detected in ground water. The use of this chemical in areas where soils are permeable, particularly where the water table is shallow, may result in ground water contamination.

This product is toxic to aquatic life with long lasting effects.

The following ecotoxicological information is for the active ingredient (imazapic):

Mallard and Bobwhite Quail (LD₅₀): > 2,150 mg/kg

Rainbow Trout and Bluegill (LC₅₀): > 100 mg/l

Daphnia (LC₅₀): > 100 mg/l

Bees (LD₅₀): > 100 µg/bee

13. DISPOSAL CONSIDERATIONS

PESTICIDE DISPOSAL: Do not contaminate water, food or feed by disposal. Wastes resulting from the use of this product (that cannot be used according to label instructions) must be disposed of on site or at an approved waste disposal facility.

CONTAINER DISPOSAL: Nonrefillable container. Do not reuse or refill this container. Triple rinse container (or equivalent) promptly after emptying. Refer to the product label for specific container handling instructions.

14. TRANSPORT INFORMATION

UN Number:	UN3082
Proper Shipping Name:	Environmentally hazardous substance, liquid, N.O.S. (contains imazapic)
Transport Hazard Class:	9
Packing Group:	III
Hazard Zone:	A
Marine Pollutant:	Yes ¹
Hazardous Substance RQ:	None
Labels / Placards:	US-DOT: Class 9 Environmentally Hazardous Substance ² IMDG, IATA: Class 9 Environmentally Hazardous Substance ²
Emergency Guide:	171 (NAERG – North American Emergency Response Guide)
¹ Marine pollutant note:	Ground-only shipments are excluded from Marine Pollutant labeling requires as per 49CFR172.101 Appendix B(4). For any shipments involving all or part of the transport by vessel, the shipment must be classified as a Marine Pollutant unless a limited quantity exemption applies (see note 3 below).
² US-DOT Note:	Not regulated for “ground only” shipments.
³ IMDG / IATA Note:	Not regulated when shipped in single or inner packaging ≤1.32 gal. (5 L).

This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Transportation classifications may vary by container volume and may be influenced by regional or country variations in regulations. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations and rules relating to the transportation of the material.

15. REGULATORY INFORMATION

FIFRA –

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. The following is the hazard information as required on the pesticide label:

PRECAUTIONARY STATEMENTS

HAZARDS TO HUMANS AND DOMESTIC ANIMALS

CAUTION. Causes moderate eye irritation. Avoid breathing spray mist. Avoid contact with skin, eyes, or clothing. Wash thoroughly with soap and water after handling and before eating, drinking, chewing gum, or using tobacco.

ENVIRONMENTAL HAZARDS

Do not apply directly to water, or to areas where surface water is present, or to intertidal areas below the mean high water mark. Do not contaminate water when cleaning equipment or disposing of equipment washwaters or rinsate. This chemical demonstrates the properties and characteristics associated with chemicals detected in ground water. The use of this chemical in areas where soils are permeable, particularly where the water table is shallow, may result in ground water contamination.

All pesticides are governed under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA). The regulatory information presented below is pertinent only when this product is handled outside of the normal use and application as a pesticide. This product is excluded from listing requirements under EPA/TSCA.

SARA Title III – Section 302 Extremely Hazardous Substances

Not listed

SARA Title III – Section 311/312 Hazard Categories

Immediate

SARA Title III – Section 312 Threshold Planning Quantity

The threshold planning quantity (TPQ) for this product treated as a mixture is 10,000 lbs. This product contains no ingredients with a TPQ of less than 10,000 lbs.

SARA Title III – Section 313 Reportable Ingredients

Not listed

CERCLA –

N/A

CALIFORNIA PROP 65 STATUS –

This product does not contain any products known to the state of California to cause cancer or reproductive harm.

CANADA –

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR.

16. OTHER INFORMATION

This Safety Data Sheet (SDS) serves different purposes than and DOES NOT REPLACE OR MODIFY THE EPA APPROVED PRODUCT LABELING (attached to and accompanying the product container). This SDS provides important health, safety, and environmental information for employers, employees, emergency responders and others handling large quantities of the product in activities generally other than product use, while the labeling provides that information specifically for product use in the ordinary course.

THIS INFORMATION IN THIS SDS IS BASED ON DATA AVAILABLE AS OF THE REVISION DATE GIVEN HEREIN, AND BELIEVED TO BE CORRECT. CONTACT ALLIGARE, LLC TO CONFIRM IF YOU HAVE THE MOST CURRENT SDS. JUDGMENTS AS TO THE SUITABILITY OF THE INFORMATION HEREIN FOR THE INDIVIDUAL'S OWN USE OR PURPOSES IS NECESSARILY THE INDIVIDUAL'S OWN RESPONSIBILITY. ALTHOUGH REASONABLE CARE HAS BEEN TAKEN IN THE PREPARATION OF SUCH INFORMATION, ALLIGARE, LLC EXTENDS NO WARRANTIES, MAKES NO

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